

Image Resolution Enhancement

Using Optimized Edge-Preserving Interpolation

Computer Vision Project Proposal

Group Members

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Problem Statement

The goal of this project is to develop an image resolution enhancement system capable of converting a low-resolution (LR) image into a high-resolution (HR) image while preserving edge sharpness and texture details. Traditional interpolation methods such as Bilinear, Bicubic, and Lanczos often introduce blurring and ringing artifacts, leading to loss of fine details.

The system takes a low-resolution image as input and produces a high-resolution output with improved sharpness, reduced artifacts, and enhanced perceptual quality. Performance will be evaluated using PSNR and SSIM metrics. The work is relevant to medical imaging, satellite imaging, surveillance, and multimedia applications.

Approach

The proposed method is based on an optimized edge-preserving interpolation framework inspired by the OLA e-spline technique.

- **Preprocessing:** Local Adaptive filtering using Unsharp Masking to enhance edges.
- **High-Frequency Extraction:** Generation of high-pass image components via subtraction.
- **Optimization:** Cuckoo Search optimization for gain tuning.
- **Interpolation:** B-spline interpolation for smooth upscaling.
- **Edge Preservation:** Edge detection-based refinement to reduce artifacts.

Implementation will be done in Python using OpenCV, NumPy, and Matplotlib.

Experiments and Results

The evaluation will use benchmark datasets such as Set5, Set14, and standard images (Lena, Barbara, Baby, Face, Airplane). Images will be downsampled using scaling factors of $\times 2$ and $\times 4$.

Compared Methods:

- Bilinear Interpolation
- Bicubic Interpolation
- Lanczos Interpolation
- Proposed Method

Evaluation Metrics:

- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index (SSIM)
- Feature Similarity Index (FSIM)

Analysis Includes:

- Edge sharpness and artifact reduction
- Visual comparison with ground truth HR images
- Computational runtime evaluation

Expected Outcome

The expected result is improved edge preservation and texture reconstruction compared to conventional interpolation methods, along with higher PSNR and SSIM values. A key challenge is balancing image quality improvement with computational complexity due to adaptive filtering and optimization steps.